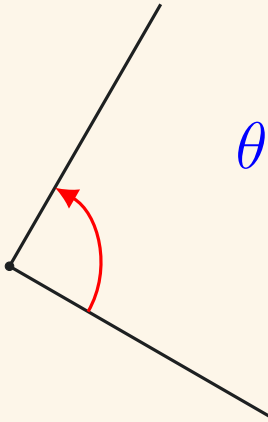


The Comprehensive List of

References in Geometry

AN ILLUSTRATED MANUAL





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Appendix A

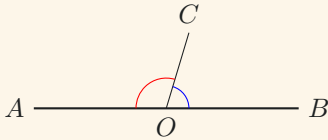
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Appendix B

Special Points and Lines in $\triangle$ s 30

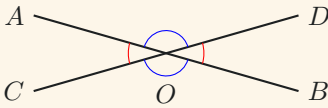
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Lines



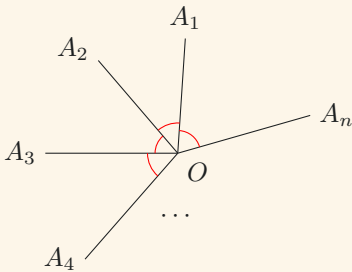
Adj. $\angle$ s on st. line

$$\angle AOC + \angle COB = 180^\circ.$$



Vert. oppo. $\angle$ s

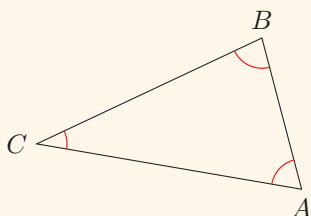
$$\begin{aligned}\angle AOC &= \angle DOB, \\ \angle AOD &= \angle COB.\end{aligned}$$



$\angle$ s at a pt.

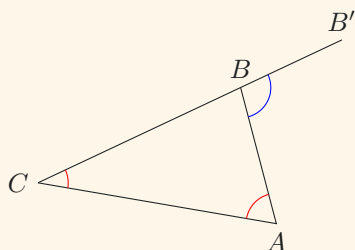
$$\begin{aligned}\angle A_1OA_2 + \angle A_2OA_3 \\ + \cdots + \angle A_{n-1}OA_n = 360^\circ.\end{aligned}$$

Triangles



$\angle$ *sum of $\triangle s$*

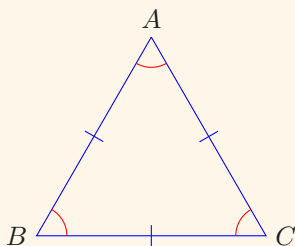
$$\angle CAB + \angle ABC + \angle BCA = 180^\circ.$$



Ext. $\angle$ *of $\triangle s$*

$$\angle CAB + \angle BCA = \angle B'BA.$$

Equilateral Triangles

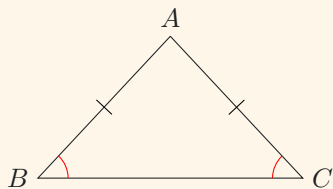


Prop. of equil. $\triangle s$

$$AB = BC = CA,$$

$$\angle CAB = \angle ABC = \angle BCA = 60^\circ.$$

Isosceles Triangles

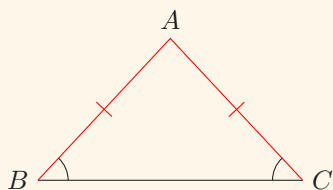


Base $\angle$ s, isos. $\triangle$

$$AB = AC$$



$$\angle ABC = \angle ACB.$$

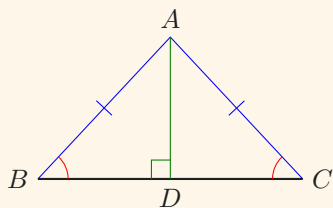


Sides oppo. eq. $\angle$ s

$$\angle ABC = \angle ACB$$



$$AB = AC.$$



Prop. of isos. $\triangle$ s

$$\angle ABC = \angle ACB$$

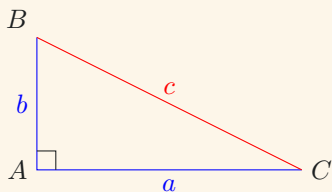


$$AB = AC$$



$$AD \perp BC.$$

Pythagoras's Theorem

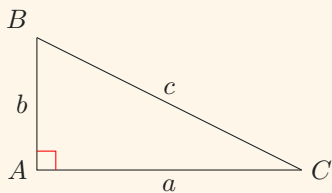


Pyth. thrm.

$$BA \perp AC$$



$$a^2 + b^2 = c^2.$$



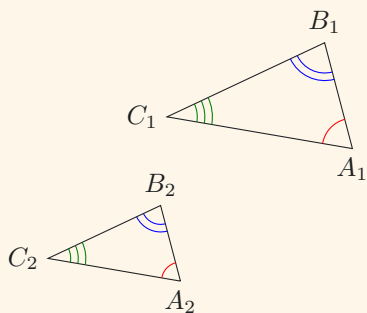
Conv. of Pyth. thrm.

$$a^2 + b^2 = c^2$$



$$BA \perp AC.$$

Similar Triangles



Corr. $\angle s \mid \sim \triangle s$

$$\triangle A_1B_1C_1 \sim \triangle A_2B_2C_2$$

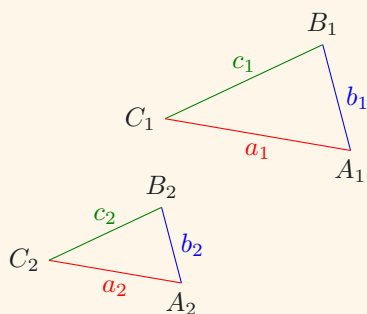


$$\angle C_1A_1B_1 = \angle C_2A_2B_2,$$

$$\angle A_1B_1C_1 = \angle A_2B_2C_2,$$

$$\angle B_1C_1A_1 = \angle B_2C_2A_2.$$

TRIANGLES



Corr. sides | $\sim \triangle s$

$$\triangle A_1B_1C_1 \sim \triangle A_2B_2C_2$$

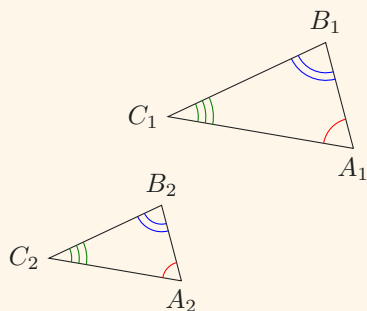
$$\Downarrow$$

$$\frac{a_1}{b_1} = \frac{a_2}{b_2},$$

$$\frac{b_1}{c_1} = \frac{b_2}{c_2},$$

$$\frac{a_1}{b_1} = \frac{a_2}{b_2}.$$

Conditions for Proving Similar Triangles



AAA

$$\angle C_1A_1B_1 = \angle C_2A_2B_2,$$

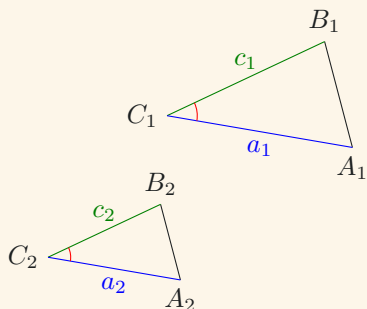
$$\angle A_1B_1C_1 = \angle A_2B_2C_2,$$

$$\angle B_1C_1A_1 = \angle B_2C_2A_2$$

$\Downarrow$

$$\underline{\underline{\triangle A_1B_1C_1 \sim \triangle A_2B_2C_2.}}$$

TRIANGLES



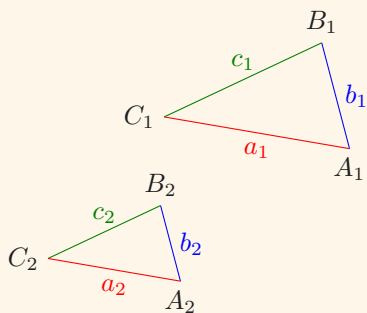
*Ratio of two sides,
incl. $\angle$*

$$\frac{a_1}{c_1} = \frac{a_2}{c_2},$$

$$\angle B_1C_1A_1 = \angle B_2C_2A_2$$



$$\underline{\underline{\triangle A_1B_1C_1 \sim \triangle A_2B_2C_2.}}$$



*Three sides
proportional*

$$\frac{a_1}{b_1} = \frac{a_2}{b_2},$$

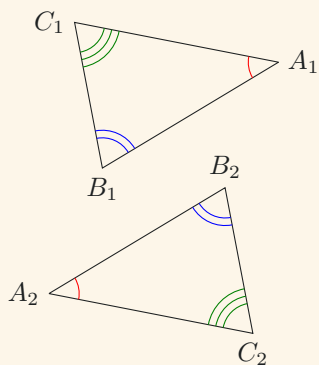
$$\frac{b_1}{c_1} = \frac{b_2}{c_2},$$

$$\frac{a_1}{b_1} = \frac{a_2}{b_2}$$



$$\underline{\underline{\triangle A_1B_1C_1 \sim \triangle A_2B_2C_2.}}$$

Congruent Triangles



Corr. $\angle s$ | $\cong \triangle s$

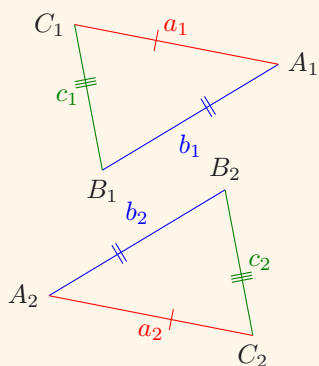
$$\triangle A_1B_1C_1 \cong \triangle A_2B_2C_2$$



$$\angle C_1A_1B_1 = \angle C_2A_2B_2,$$

$$\angle A_1B_1C_1 = \angle A_2B_2C_2,$$

$$\angle B_1C_1A_1 = \angle B_2C_2A_2.$$



Corr. sides | $\cong \triangle s$

$$\triangle A_1B_1C_1 \cong \triangle A_2B_2C_2$$



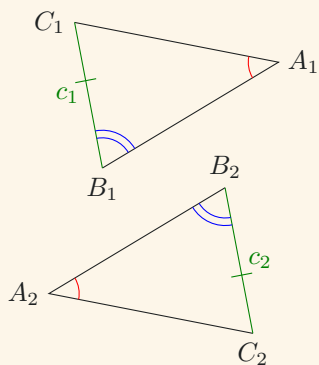
$$a_1 = a_2,$$

$$b_1 = b_2,$$

$$c_1 = c_2.$$

TRIANGLES

Conditions for Proving Congruent Triangles



AAS

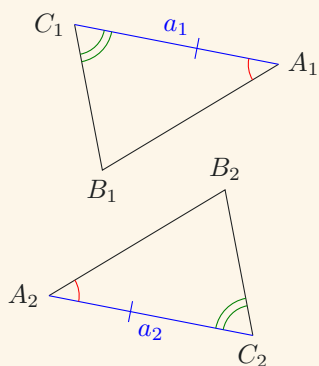
$$\angle C_1A_1B_1 = \angle C_2A_2B_2,$$

$$\angle A_1B_1C_1 = \angle A_2B_2C_2,$$

$$c_1 = c_2$$



$$\underline{\underline{\triangle A_1B_1C_1 \cong \triangle A_2B_2C_2.}}$$



ASA

$$\angle A_1B_1C_1 = \angle C_2A_2B_2,$$

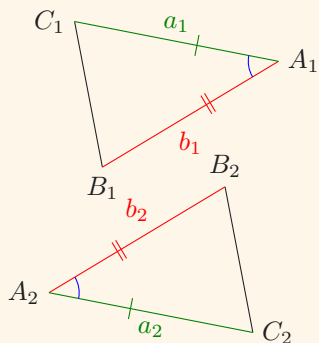
$$a_1 = a_2,$$

$$\angle B_1C_1A_1 = \angle B_2C_2A_2,$$



$$\underline{\underline{\triangle A_1B_1C_1 \cong \triangle A_2B_2C_2.}}$$

TRIANGLES

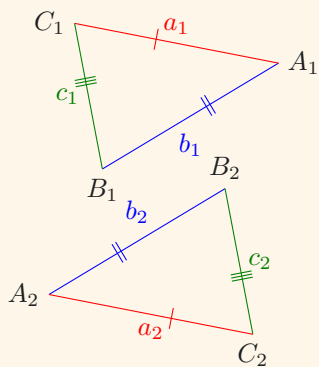


SAS

$$\begin{aligned} a_1 &= b_1, \\ \angle B_1 A_1 C_1 &= \angle B_2 A_2 C_2, \\ b_1 &= b_2 \end{aligned}$$



$$\underline{\underline{\triangle A_1 B_1 C_1 \cong \triangle A_2 B_2 C_2.}}$$



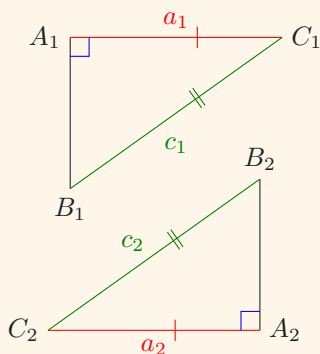
SSS

$$\begin{aligned} a_1 &= a_2, \\ b_1 &= a_2, \\ c_1 &= a_2 \end{aligned}$$



$$\underline{\underline{\triangle A_1 B_1 C_1 \cong \triangle A_2 B_2 C_2.}}$$

POLYGONS



RHS

$$\begin{aligned}\angle B_1A_1C_1 &= \angle B_2A_2C_2 \\ &= 90^\circ,\end{aligned}$$

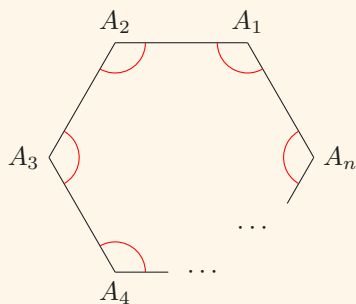
$$c_1 = c_2,$$

$$a_1 = a_2$$



$$\underline{\underline{\triangle A_1B_1C_1 \cong \triangle A_2B_2C_2.}}$$

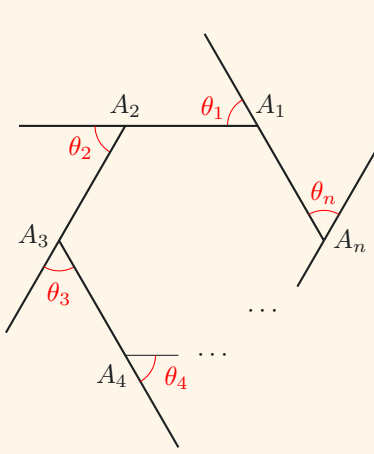
Polygons



$\angle$ *sum of polygons*

$$\begin{aligned}\angle A_nA_1A_2 + \angle A_1A_2A_3 + \dots \\ + \angle A_{n-1}A_nA_1 = (n-2) \cdot 180^\circ.\end{aligned}$$

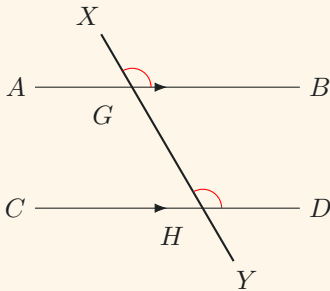
PARALLEL LINES



Ext. $\angle$ s of polygons

$$\theta_1 + \theta_2 + \cdots + \theta_n = 360^\circ.$$

Parallel Lines



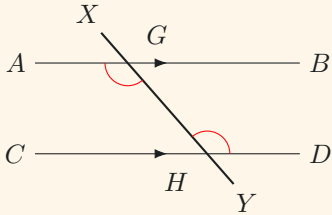
Corr. $\angle$ s | $AB \parallel CD$

$$AB \parallel CD$$



$$\angle XGB = \angle XHD.$$

PARALLEL LINES

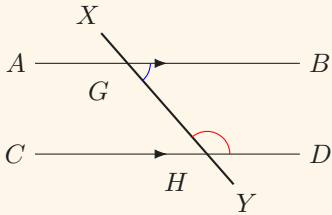


Alt. $\angle s$ | $AB \parallel CD$

$AB \parallel CD$



$\angle YGA = \angle XHD.$



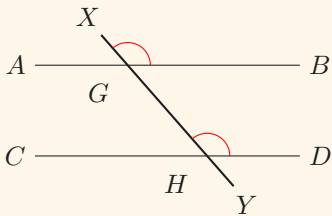
Int. $\angle s$ | $AB \parallel CD$

$AB \parallel CD$



$\angle YGB + \angle XHD = 180^\circ.$

Conditions for Proving Parallelism



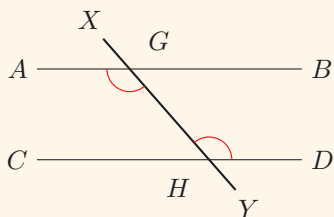
Int. $\angle s$ eq.

$\angle YGB + \angle XHD = 180^\circ$



$AB \parallel CD.$

PARALLELOGRAMS

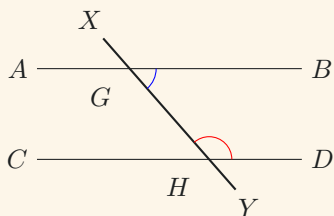


Alt. $\angle$ s eq.

$$\angle YGA = \angle XHD$$



$$\underline{\underline{AB \parallel CD.}}$$



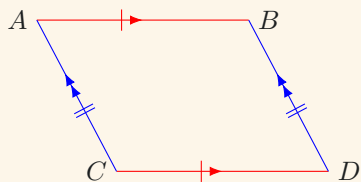
Int. $\angle$ s eq

$$\angle YGB + \angle XHD = 180^\circ$$



$$\underline{\underline{AB \parallel CD.}}$$

Parallelograms

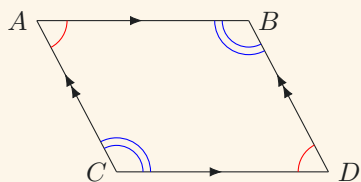


Oppo. sides of $\parallel$ -gram

$$AB = CD,$$

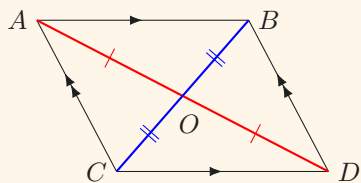
$$AC = BD.$$

PARALLELOGRAMS



Oppo. $\angle$ s of $\parallel$ -gram

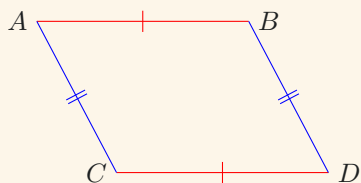
$$\begin{aligned}\angle BAC &= \angle BDC, \\ \angle ACD &= \angle ABD.\end{aligned}$$



Diags. of $\parallel$ -gram

$$\begin{aligned}OA &= OD, \\ OB &= OC.\end{aligned}$$

Conditions for Identifying Parallelograms



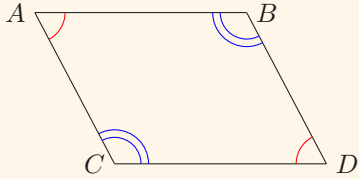
Oppo. sides eq.

$$\begin{aligned}AB &= CD, \\ AC &= BD\end{aligned}$$



$ABCD$ is a $\parallel$ -gram.

PARALLELOGRAMS



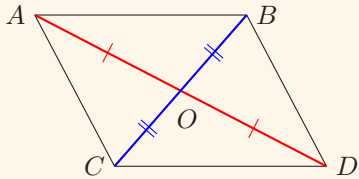
Oppo. $\angle$ s eq.

$$\angle BAC = \angle BDC,$$

$$\angle ACD = \angle ABD$$



$ABCD$ is a $\parallel$ -gram.



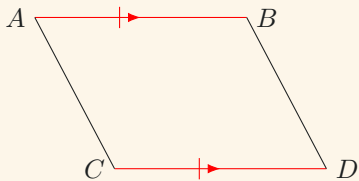
Diags. bisect each other

$$OA = OD,$$

$$OB = OC$$



$ABCD$ is a $\parallel$ -gram.



Two sides eq. and $\parallel$

$$AB = CD,$$

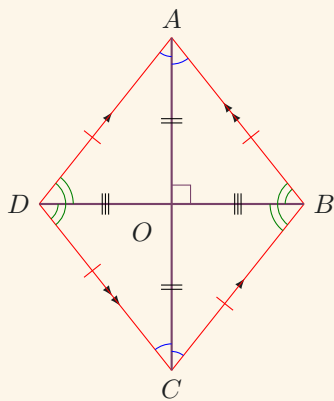
$$AB \parallel CD$$



$ABCD$ is a $\parallel$ -gram.

Other Quadrilaterals

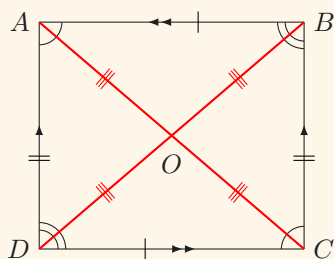
Rhombus - a quadrilateral with four equal sides



Prop. of rhombuses

$$\begin{aligned}
 AB &\parallel CD & AD &\parallel BC, \\
 OA &= OC & OB &= OD, \\
 AB &= BC = CD = DA, \\
 AC &\perp BD, \\
 \angle OAD &= \angle OAB \\
 &= \angle OCD = \angle OCB, \\
 \angle ODA &= \angle ODC \\
 &= \angle OBA = \angle OBC.
 \end{aligned}$$

Rectangle - a quadrilateral with four equal interior angles

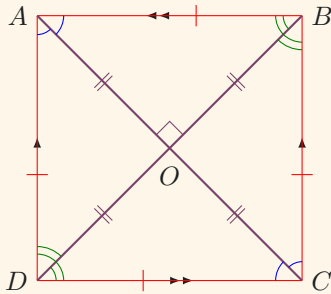


Prop. of rectangles

$$\begin{aligned}
 AB &\parallel CD & AD &\parallel BC, \\
 AB &= CD & AD &= BC, \\
 \angle BAD &= \angle BCD, \\
 \angle ABC &= \angle ADC, \\
 OA &= OB = OC = OD.
 \end{aligned}$$

QUADRILATERALS

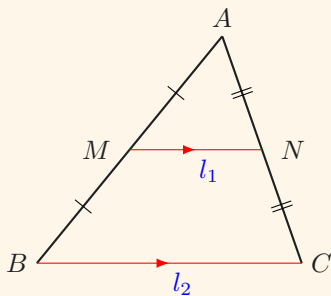
Square - a quadrilateral with four equal sides and interior angles



Prop. of squares

$$\begin{aligned}
 &AB \parallel CD \quad AD \parallel BC, \\
 &AB = BC = CD = DA, \\
 &OA = OB = OC = OD, \\
 &AC \perp BD, \\
 &\angle OAD = \angle OAB = 45^\circ \\
 &\quad = \angle OCD = \angle OCB, \\
 &\angle ODA = \angle ODC = 45^\circ \\
 &\quad = \angle OBA = \angle OBC.
 \end{aligned}$$

Miscellaneous Results



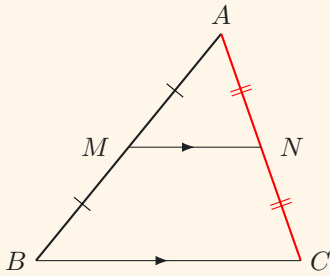
Mid-pt. thrm.

$$MA = MB \quad NA = NC$$



$$\begin{aligned}
 &2l_1 = l_2, \\
 &MN \parallel BC.
 \end{aligned}$$

QUADRILATERALS



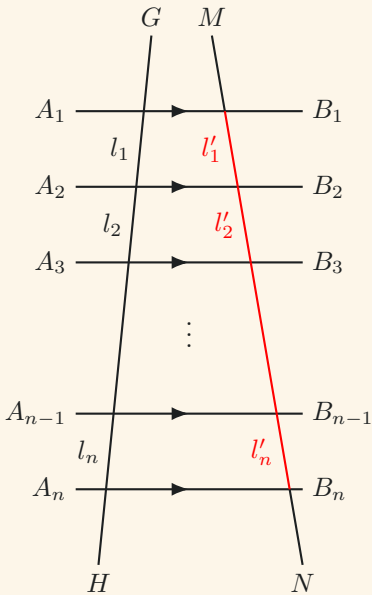
Intercept thrm.

variant i

$$MA = NA \quad MN \parallel BC$$



$$MA = NA.$$



Intercept thrm.

variant ii

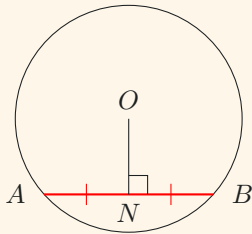
$$A_1B_1 \parallel A_2B_2 \parallel \dots \parallel A_nB_n$$

$$l_1 = l_2 = \dots = l_n$$



$$l'_1 = l'_2 = \dots = l'_n.$$

Circles

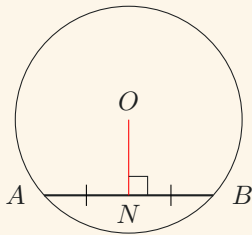


*Line from cen. $\perp$
chord bisects chord*

$$ON \perp AB$$



$$NA = NB.$$



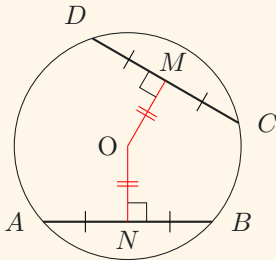
*Line joining cen.
to mid-pt. of
chord $\perp$ chord*

$$NA = NB$$



$$ON \perp AB.$$

CIRCLES



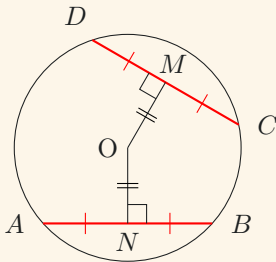
*Eq. chords
equidis. from cen.*

$$ON \perp AB \quad OM \perp CD,$$

$$AB = CD$$



$$ON = OM.$$



*Chords equidis. from
cen. are eq.*

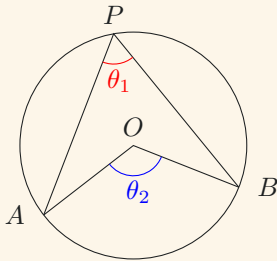
$$ON \perp AB \quad OM \perp CD,$$

$$ON = OM$$



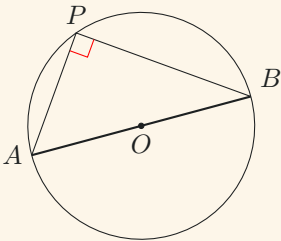
$$AB = CD.$$

CIRCLES



$\angle$ at cen. twice
 $\angle$ at $\odot^{ce}$

$$2\theta_1 = \theta_2.$$

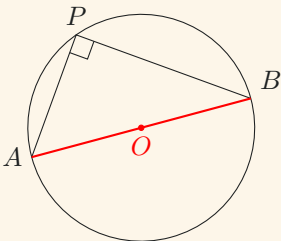


$\angle$ in semi-circ.

AB is a diameter



$$\angle APB = 90^\circ.$$



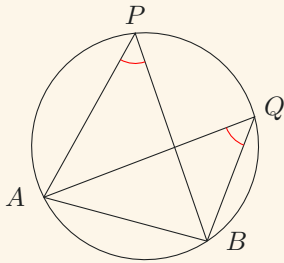
Converse of
 $\angle$ in semi-circ.

$$\angle APB = 90^\circ$$



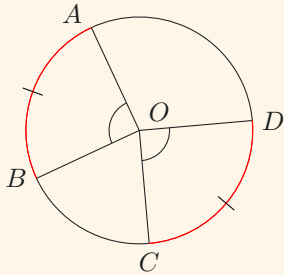
AB is a diameter.

CIRCLES



$\angle s$ in the same segment

$$\angle APB = \angle AQB.$$

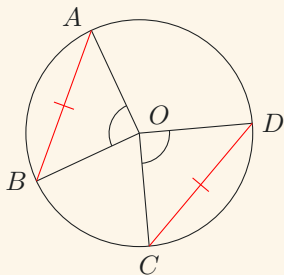


Eq. $\angle s$, eq. arcs

$$\angle AOB = \angle COD$$



$$\widehat{AB} = \widehat{CD}.$$



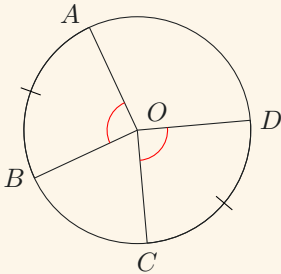
Eq. $\angle s$, eq. chords

$$\angle AOB = \angle COD$$



$$AB = CD.$$

CIRCLES

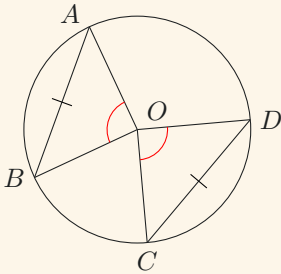


Eq. arcs, eq. $\angle$ s

$$\widehat{AB} = \widehat{CD}$$



$$\angle AOB = \angle COD.$$

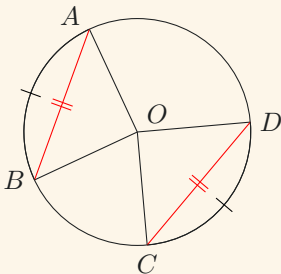


Eq. chords, eq. $\angle$ s

$$AB = CD$$



$$\angle AOB = \angle COD.$$



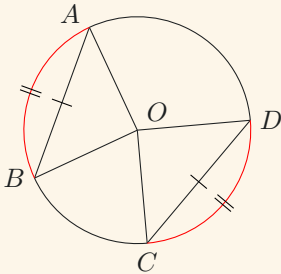
Eq. arcs, eq. chords

$$\widehat{AB} = \widehat{CD}$$



$$AB = CD.$$

CIRCLES

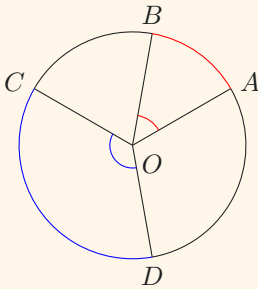


Eq. chords, eq. arcs

$$AB = CD$$

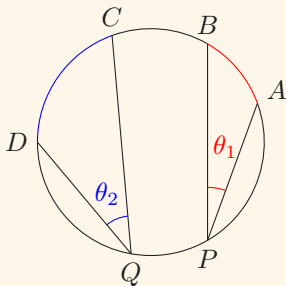


$$\widehat{AB} = \widehat{CD}.$$



Arcs prop. to $\angle$ s at cen.

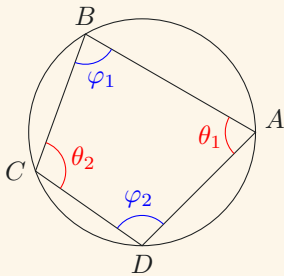
$$\frac{\widehat{AB}}{\widehat{CD}} = \frac{\theta_1}{\theta_2}.$$



Arcs prop. to $\angle$ s at $\odot^{ce}$

$$\frac{\widehat{AB}}{\widehat{CD}} = \frac{\theta_1}{\theta_2}.$$

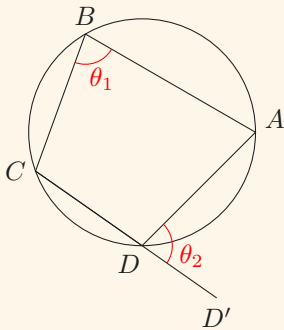
Cyclic Quadrilaterals



Oppo. $\angle$ s, cyclic quad.

$$\theta_1 + \theta_2 = 180^\circ,$$

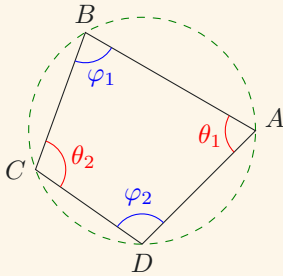
$$\varphi_1 + \varphi_2 = 180^\circ.$$



Ext. $\angle$ s, cyclic quad.

$$\theta_1 = \theta_2.$$

Conditions for Identifying Cyclic Quadrilaterals



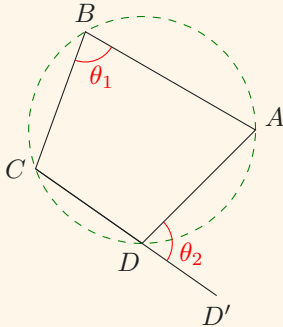
Oppo. $\angle$ s supp.

$$\theta_1 + \theta_2 = 180^\circ,$$

$$\varphi_1 + \varphi_2 = 180^\circ$$



A, B, C and D are concyclic.



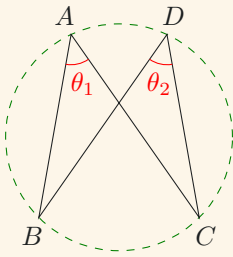
Ext. $\angle$ = int. oppo. $\angle$

$$\theta_1 = \theta_2$$



A, B, C and D are concyclic.

CIRCLES



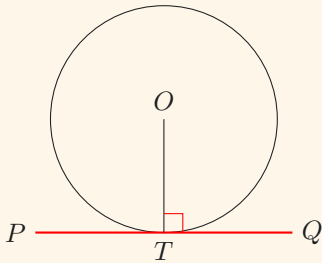
*Converse of $\angle s$
in the same segment*

$$\theta_1 = \theta_2$$



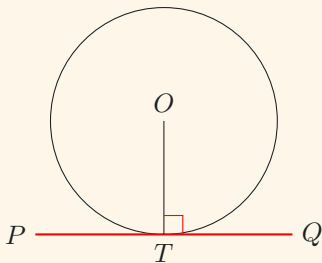
A, B, C and D are concyclic.

Tangents



Tan. $\perp$ radius

$$OT \perp PQ.$$



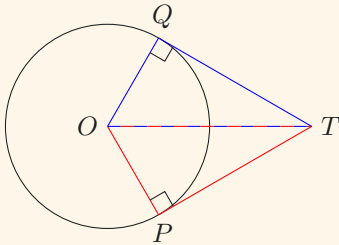
*Converse of
tan. $\perp$ radius*

$$OT \perp PQ.$$



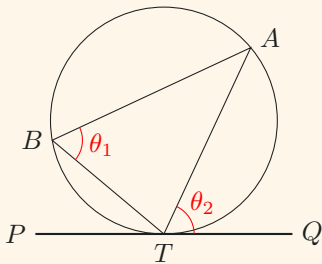
PQ is tangent
to the circle at T .

CIRCLES



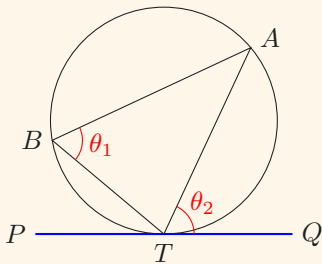
Tan. prop.

$$\triangle OPT \cong \triangle OQT.$$



∠ in alt. segment

$$\theta_1 = \theta_2.$$



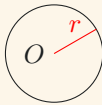
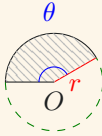
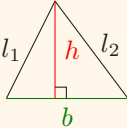
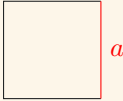
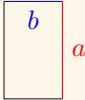
*Converse of
∠ in alt. segment*

$$\theta_1 = \theta_2$$

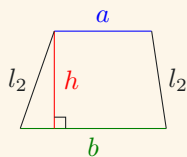


PQ is tangent
to the circle at *T*.

Appendix A: Area and Perimeters of Common Plane Figures

FIGURE	AREA	PERIMETER	
	πr^2	$2\pi r$	◀ CIRCLES
	$\frac{\pi \theta r^2}{360^\circ}$	$\frac{2\pi \theta r}{360^\circ} + 2r$	◀ SECTORS
	$\frac{hb}{2}$	$b + l_1 + l_2$	◀ TRIANGLES
	a^2	$4a$	◀ SQUARES
	ab	$2(a + b)$	◀ RECTANGLES

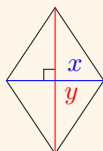
SPECIAL POINTS AND LINES IN $\triangle$ S



$$\frac{h(a+b)}{2}$$

$$a+b+l_1+l_2$$

◀ TRAPEZIUMS



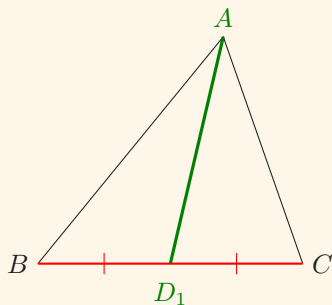
$$\frac{xy}{2}$$

$$2\sqrt{x^2+y^2}$$

◀ RHOMBUSES

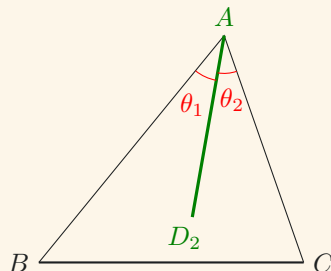
Appendix B: Special Points and Lines in $\triangle$ s

ΩAD_1 is the *median* of BC



$$BD_1 = CD_1.$$

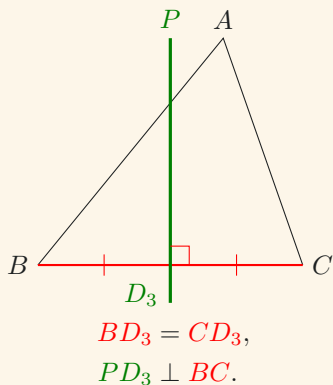
ΩAD_2 is an *angle bisector* of $\angle BAC$



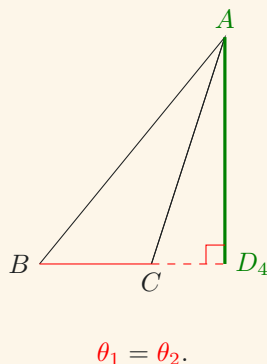
$$\theta_1 = \theta_2.$$

SPECIAL POINTS AND LINES IN $\triangle$ s

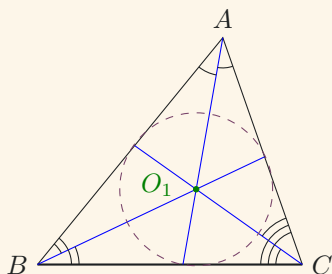
ΩPD_3 is a *perpendicular bisector* of BC



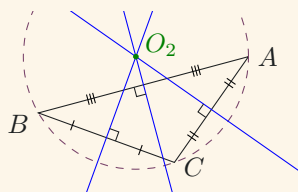
ΩAD_2 is the *altitude* of BC



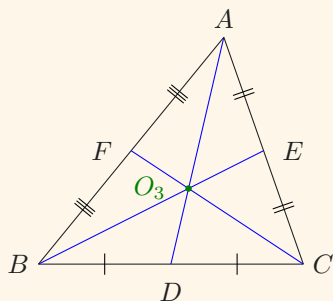
ΩO_1 is the *in-centre* of $\triangle ABC$



ΩO_2 is the *circumcentre* of $\angle BAC$



ΩO_1 is the *centroid* of $\triangle ABC$

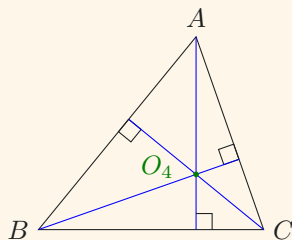


The *centroid* is the point of intersection of the three *medians* (of a triangle).

It divides each median in the ratio of 2 : 1, i.e.,

$$\frac{O_3A}{O_3D} = \frac{O_3B}{O_3E} = \frac{O_3C}{O_3F} = 2.$$

ΩO_4 is the *orthocentre* of $\triangle ABC$



The *orthocentre* is the point of intersection of the three *altitudes*.

These four points are generally distinct from one another.